

Task 1 – Communication System

For this, my priorities would be:

1. **Reliability, precision and all-time performance:** Out in the sea on a search-and-rescue mission, there is no room for screw-ups or mistakes. Thus, I would put this as my first parameter (even in general, I feel it is better to spend more for reliability, which eventually pays off).
2. **Weight:** As we have been given a strict weight requirement, which we will aim to minimize for maximum range, I opt for this as my second parameter.
3. **Compute:** Safe to say, the drone will face a very chaotic and confusing situation in front of it. To ensure it can safely and properly navigate and manage this data, I would put compute as my next priority.
4. **Power:** This one is bit of a balancing act – on one hand, high compute necessitates more power, while low weight requires smaller batteries, and so, less power. As a result of its being controlled by these two factors, I put this factor here.
5. **Cost:** We are trying to save people (and not money) here – besides, we can later repurpose the drone for other tasks, and recoup the losses, which cannot be guaranteed by a cheaper, less reliable drone. Thus, I put this factor dead last

Based on these factors I choose the **LiDAR + Raspberry Pi 5** combination for my drone (by elimination - stereo cameras are purely dependent on computer vision/AI for distance and object recognition, risking inaccuracy, depth cameras have issues in less-than-ideal lighting – but also, the Pi's lower cost and vis-a-vis the Jetson and OAK D-LITE, and its well-established amateur and hobby programming ecosystem, and lower overall weight due to lesser cooling).

Bonus Task

For the sensor orientation, I would opt for 3 on the front - 1 facing head-on (frontal obstacle detection -wide angle), 2 facing a little to the sides (approaching obstacle detection) – 4 on each propellor arm (side obstacle detection) and 1 (the wide angle – also on the front centre) on the back (rear obstacle detection).

AS an alternative, I would take the **Intel DepthSense D435i + Jetson Orin** combo (or add that as a supplementary sensor set to my current one – depth cameras give proper values with minimal scope of error, while the Jetson can handle the full object detection and avoidance routine on its NPU/GPU).